ENTRUSTing WfExS 1.0 with WES and TES

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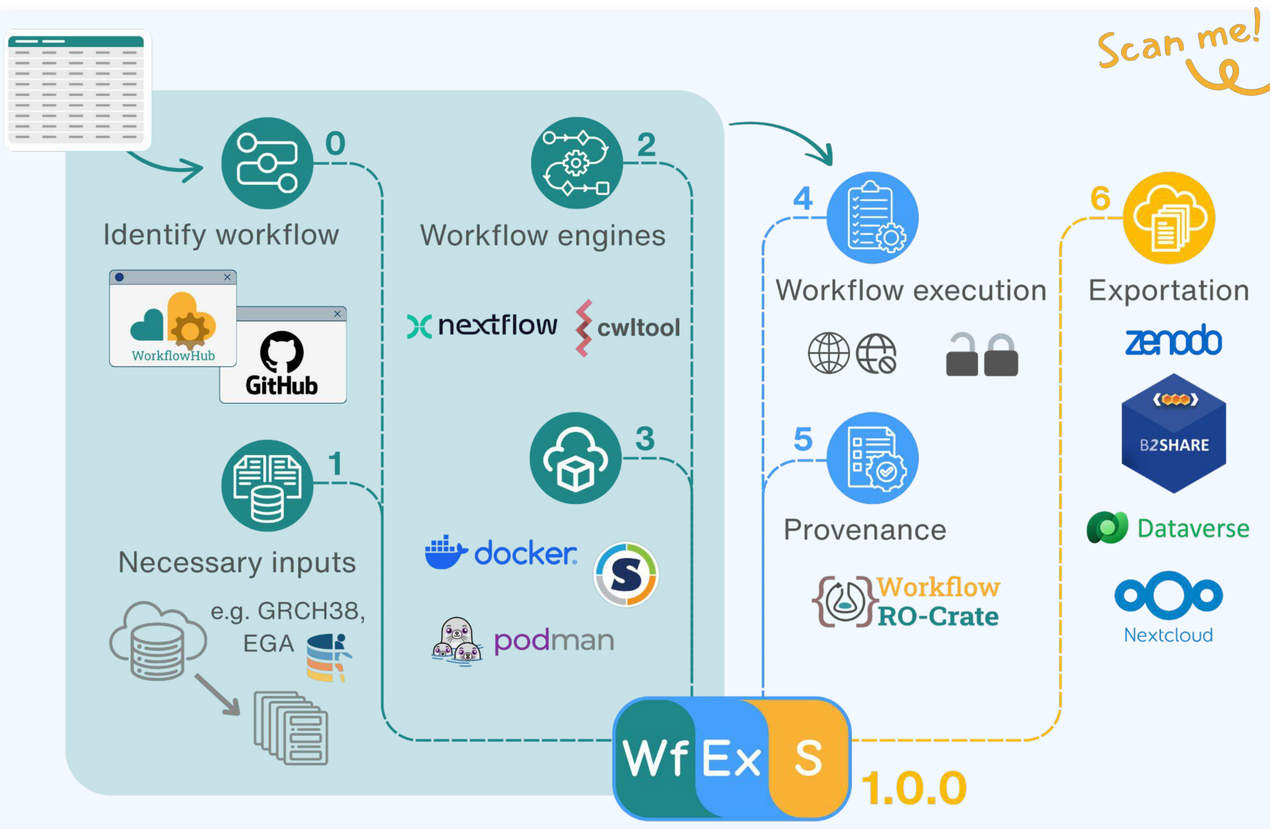
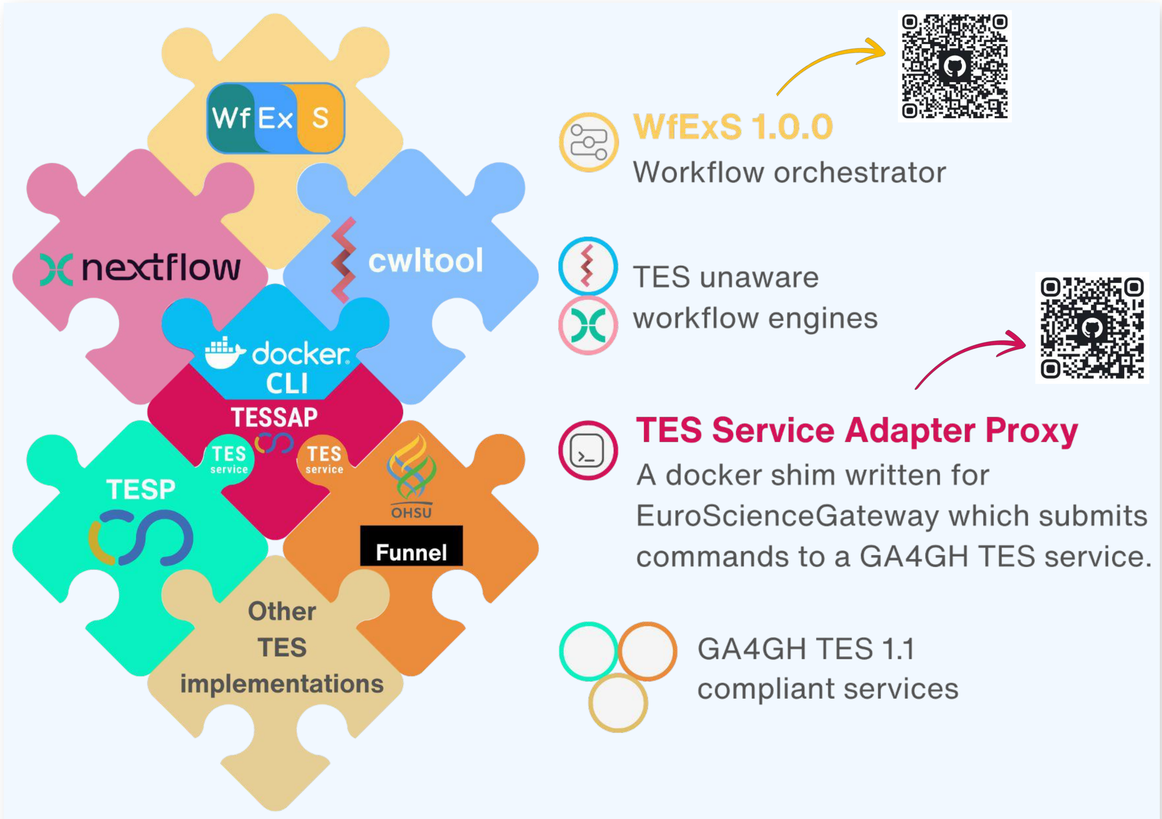
Scan me!



WfExS-backend 1.0.0

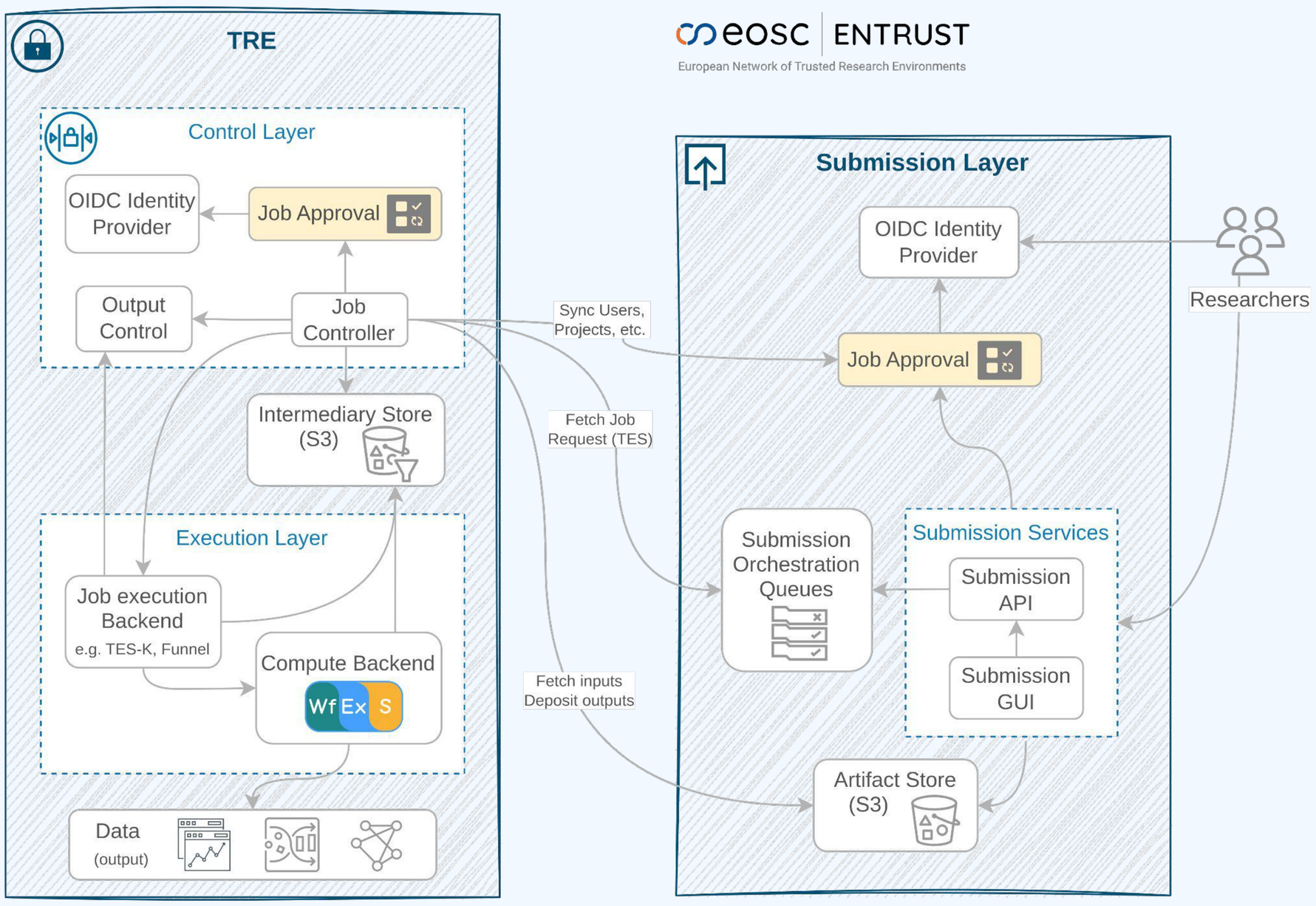
- High-level workflow engine orchestrator.
- Secure and reproducible workflow executions: Isolated Containerised
- Delegates workflow execution to existing workflow engines.
- Clear separation: resource gathering, workflow instantiation and secure result export.
- Workflow execution provenance: Workflow Run RO-Crate.
- Execution results can be exported to either private or public deposition sites.



TES Service Adapter Proxy (TESSAP)

WfExS-backend delegates workflow execution to the appropriate existing workflow engines — currently, Nextflow or cwltool. But not all the workflow engines support standards like GA4GH TES equally.

- Instead of relying on each engine's TES support, we leverage a shared strength: **TESSAP**
- By intercepting and replacing the docker command, we maintain CLI compatibility
- and enable smooth, engine-agnostic workflow execution!



WfExS-backend features in a TRE

	WfExS	TRE
Input fetching	⊘	✓
Workflow fetching	⊘	✓
Container fetching	⊘	✓
Workflow engine fetching	⊘	✓
External job control	⊘	✓
Internal job control	✓	✓†
Provenance	✓	✓‡
Output export	⊘	✓
Safe working directories	✓	✓
Job distribution in a federation	⊘	✓

† TRE allows the execution and can finalize the jobs.
‡ TRE gathers the provenance outside the workflow analysis.

Contact & Funding

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